

Adaptive Virtual Environments in Small Physical Spaces: Navigation Design for Customized User Experiences in Virtual Reality

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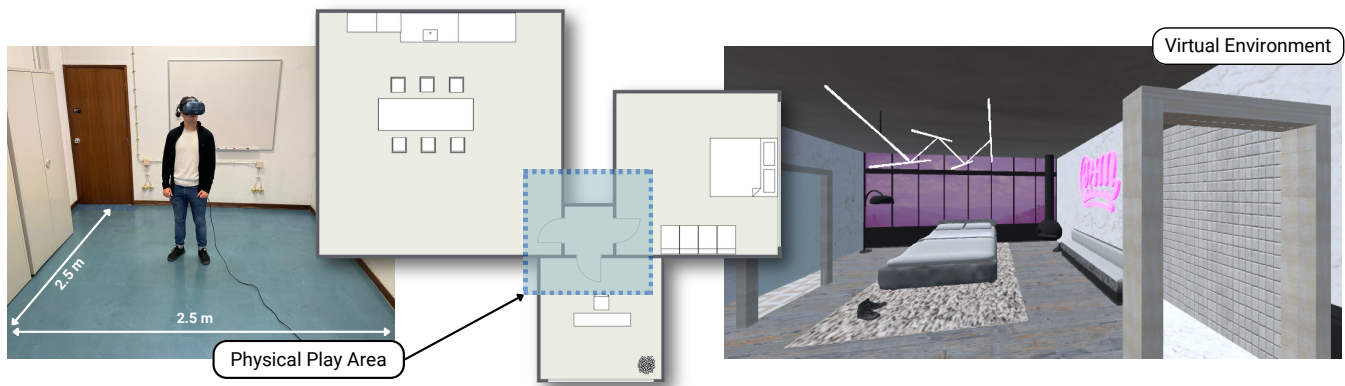


Figure 1: Our solution is a tool that either adapts a given virtual environment or generates a new one from scratch to fit into a small playing area, enabling more natural navigation without the need for handheld controllers.

ABSTRACT

In Virtual Reality (VR), navigating within small physical spaces often relies on handheld controller methods such as teleportation and joystick movements. While effective, these methods diverge from natural movement. This paper introduces the development of a VR tool designed to enhance immersion by customizing virtual environments for small physical spaces. By employing non-Euclidean and impossible spaces strategies, our solution enables overlapping virtual environments to fit the user's physical constraints, allowing exploration of the space through walking. This tool can benefit different areas of application, such as virtual visits to museums, training simulations and emergency scenarios. As a use case in this paper, we address virtual apartment tours. This study demonstrates the potential for more user-centric and accessible VR solutions, contributing to delivering more meaningful experiences in daily and professional settings. To assess the viability and potential applications of our approach, we conducted semi-structured interviews with 10 Human-Computer Interaction (HCI) researchers. These

interviews yielded insights into practical applications, identified usability challenges, and suggested strategies to enhance user experience.

CCS CONCEPTS

• **Human-centered computing** → *Interaction techniques*; **Human computer interaction (HCI)**; • **Computing methodologies** → **Virtual reality**.

KEYWORDS

Virtual reality, Non-Euclidean spaces, Impossible spaces, Navigation, Procedural generation

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1 INTRODUCTION

Virtual Reality (VR) promises immersive experiences, yet its potential is often limited by the dimensions of the physical play area. Typically, the uncluttered play area available to VR users at home

does not exceed 2x2 meters [1, 23]. In these cases, the most commonly used VR navigation techniques rely on handheld controllers for teleportation and joystick movements, compromising the natural sensation of walking and diminishing the overall sense of immersion [10, 18]. Even advanced techniques such as redirected walking [15], which rely on subtly adjusting users' visual perception to keep them away from physical boundaries, face challenges in very small areas [1]. These challenges can lead to noticeable adjustments that may induce disorientation or motion sickness, underscoring the critical need for innovative solutions tailored to constrained physical environments.

We propose a tool that aims to enhance navigation within these limitations (Figure 1). We are using non-Euclidean geometry [27], such as the concept of impossible spaces [7, 22], where the layout of the space does not conform to traditional geometric principles. For example, a corridor might appear longer on the inside than it seems from the outside, or two spaces that should be separate may connect seamlessly. By employing these techniques, our tool enables the seamless coexistence and overlap of multiple virtual spaces within the same physical space [24]. This is achieved by combining different strategies, such as portals and corridors, to naturally guide users away from the physical play area boundaries. This method not only maximizes the use of available physical space but also significantly enhances the sense of immersion and natural movement in VR, even in small spaces, by allowing users to walk without the need for handheld controllers. Virtual museum tours, virtual shopping in the metaverse and even gaming can benefit greatly from this tool, which offers a much more engaging experience tailored to the limited play areas users usually have at home.

As a use case for this study, we selected virtual apartment tours as an example that can benefit the real estate industry by enabling remote visits to properties. Our tool is capable of receiving an existing layout, for example, a house that is for sale, or procedurally generating [9, 11, 19] a layout from scratch according to predefined rules. This procedural generation is useful for simulations requiring various training scenarios, such as firefighter simulations. With procedural generation, our tool can create overlapping virtual environments by applying strategies that align with user experience requirements and the constraints of their actual physical surroundings.

We contribute a formative evaluation with 10 HCI researchers to elicit research directions for impossible spaces based on an implementation of different well-established techniques. This paper introduces our solution and the progress we have made in creating a more space-efficient VR experience. Our goal is to enable users to enjoy rich, personalized VR experiences that feel natural and intuitive, even in limited physical spaces.

2 RELATED WORK

In the domain of VR navigation constrained by limited physical spaces, researchers have explored unconventional spatial concepts such as non-Euclidean geometries [13, 16, 17, 27] and their specific application in creating impossible spaces [6, 12, 22]. These approaches focus on strategies such as designing intricate corridors [25] and employing portals or transitional areas [21, 22] to

transcend the limitations of physical space. Non-Euclidean geometries provide a foundation for these strategies, enabling VR environments to exhibit properties such as curved spaces and complex spatial configurations [8, 17]. Complementing these are flexible spaces [26], which integrate space overlapping with dynamic layout generation to create corridors and user paths.

While these techniques effectively merge multiple virtual spaces into a single physical area, selecting a single strategy for specific scenarios remains a complex challenge. Furthermore, relying on just one approach might not fully address all requirements. The key to creating a perfectly customized experience could lie in the effective integration of various elements, such as corridors, portals, rooms, and others, to meet both the user's needs and the context of use. This process requires a careful balance between maintaining spatial coherence and ensuring ease of navigation, while also minimizing potential discomfort for users.

Employing non-Euclidean geometry and impossible spaces typically requires a manual and time-consuming process to achieve a coherent layout. To streamline this, Procedural Content Generation (PCG) offers algorithms that aid in the creation of virtual environments, allowing designers to simply specify the desired characteristics of these environments [3, 9, 11]. These algorithms boost efficiency and creativity in content creation and are extensively utilized across various fields, including computing, textures, models, simulations, and video games [4, 5].

Two algorithms, Delaunay Triangulation and Minimum Spanning Tree, are particularly relevant for virtual environment layouts. Delaunay Triangulation [14, 20] involves creating a triangulation from a set of points in a plane, ensuring that no point lies inside the circumcircle of any triangle. This is crucial in fields such as terrain modelling, as it allows for flexible resolution adjustment based on application needs. The Minimum Spanning Tree [2] focuses on creating a tree with the least total weight that connects all vertices in a weighted, undirected, and connected graph, which is vital for optimizing virtual environments. These concepts together will be particularly useful to the virtual layout generation.

3 ADAPTIVE VIRTUAL ENVIRONMENTS IN SMALL PHYSICAL SPACES

Our solution presents a tool designed to either adapt an existing virtual environment or create a new one, ensuring it fits within the available physical play area. The result is a personalized navigation experience tailored to each unique application scenario. This adaptation involves segmenting the virtual layout (e.g., a house) into manageable cells (e.g., rooms), strategically arranging them to ensure seamless navigation within the limited physical space. This spatial arrangement employs non-Euclidean concepts, including portals, corridors, adjustment of the user's virtual stride, and other strategies, to determine the most suitable techniques for accommodating and connecting the rooms.

3.1 Virtual Environment Customization

The algorithm responsible for arranging or generating virtual environments operates on a comprehensive set of parameters. These parameters address both the physical and virtual world's constraints,

including the dimensions of the available play area, the intended virtual environment size, and the desired level of realism. This realism parameter significantly influences the system’s decision-making process by dictating the extent to which the algorithm can modify the space layout. A higher realism level ensures the virtual environment closely mirrors the initial layout or real-world elements, limiting the algorithm’s freedom to create noticeable impossible spaces. On the other hand, a lower level of realism allows the algorithm greater latitude to reconfigure the space, potentially diverging significantly from the original layout or real-world physics and aesthetics. Within this realism parameter, sub-parameters guide the algorithm in layout design and room connection, each uniquely affecting the user experience. While we aim to broaden our range of strategies, in the current state of development we are employing techniques including portals, corridors, interconnected rooms, and stride adjustments based on the application scenario. For instance, a science fiction virtual environment aiming for a surreal ambiance might heavily utilize futuristic portals to create fantastical transitions between spaces. Conversely, a virtual environment simulating the interior of a building might focus on interconnected rooms and corridors to achieve a realistic representation. The system offers preset configurations for commonly encountered scenarios, in addition to the option for custom presets. This feature allows users to prioritize specific navigation techniques or omit others, ensuring the virtual environment aligns precisely with the unique requirements of varied application scenarios.

3.2 Virtual Environment Construction and Integration

Figure 2 presents a simplified illustration of the main steps our tool undergoes to adapt or generate a virtual layout for the available play area. In step (A), the user defines the parameters. If a virtual layout is provided, step (B) is skipped, as generating a new layout becomes unnecessary. Otherwise, step (B) involves generating a diverse array of virtual rooms using a procedural generation algorithm to create rooms of various sizes and shapes. Key rooms are identified, and their central points (marked by circles in the figure) undergo a Delaunay Triangulation process (indicated by lines) to form a graph representing the virtual environment. This graph is essential for mapping room connections and pathways. Subsequently, a minimum spanning tree (MST) is generated within

this graph. In the figure, the full lines indicate the path chosen by the MST, ensuring an efficient route that connects the most distant rooms and provides accessibility to all areas of the virtual space. In step (C), the final virtual environment layout is displayed, showcasing all rooms and connections. Step (D) then focuses on integrating this virtual layout into the user’s physical environment. The system begins by selecting a starting room and evaluating adjacent rooms for overlap, making adjustments to fit the physical space’s constraints. The choice of connection techniques between rooms is determined by a scoring system, that is heavily influenced by the initial parameters, particularly the realism parameter. For example, in scenarios requiring high realism, corridors are preferred over more imaginative elements such as portals. Each selected technique is carefully integrated to ensure a smooth and safe transition between rooms. This step is critical, as each connection method presents unique implementation challenges that must be meticulously addressed to preserve the integrity and navigability of the virtual environment. Step (E) presents the final result, with all necessary adjustments made to the virtual layout to ensure its compatibility with the physical space.

3.3 Case Study: Virtual Apartment Tours

In our case study, we focused on the application of virtual apartment tours, a concept gaining traction in the real estate sector. The challenge involved adapting a house of around 100 m² into a 6.25 m² physical play area. The following examples aim to demonstrate different spatial arrangements that the algorithm can generate. The initial parameters were kept simple and included a virtual apartment layout, an exclusive preference for one of the strategies (i.e., portals, corridors, interconnecting rooms, or limited walking), and a moderate level of realism. This level of realism allows connecting rooms that were not originally linked, facilitating more fluid navigation without requiring the user to move back and forth between rooms. In conjunction with the following strategies, we adjusted the user’s stride in the virtual world for rooms larger than the physical space, enabling complete user access to the virtual area. The selected examples demonstrate the individual use of four strategies (Figure 3), yet combining them is recommended to achieve a more tailored solution and harness the unique benefits each strategy offers.

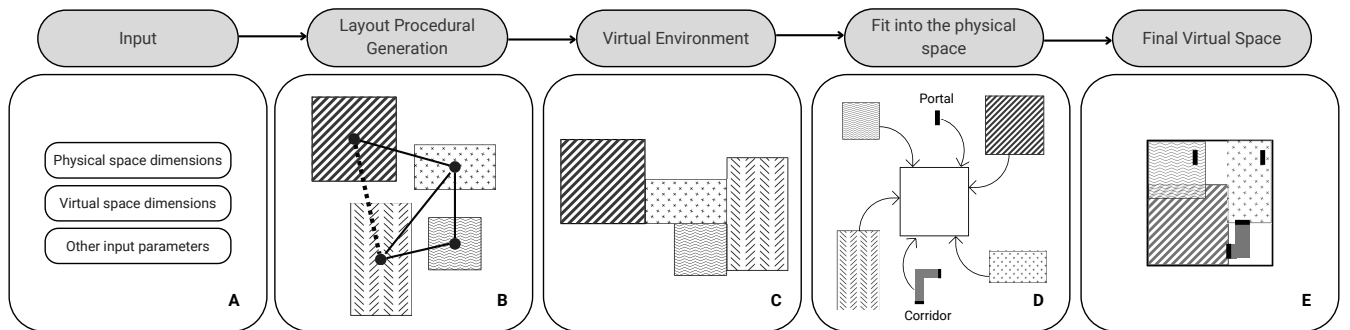


Figure 2: Overview of the tool’s steps to adapt or generate a virtual environment for the available play area.

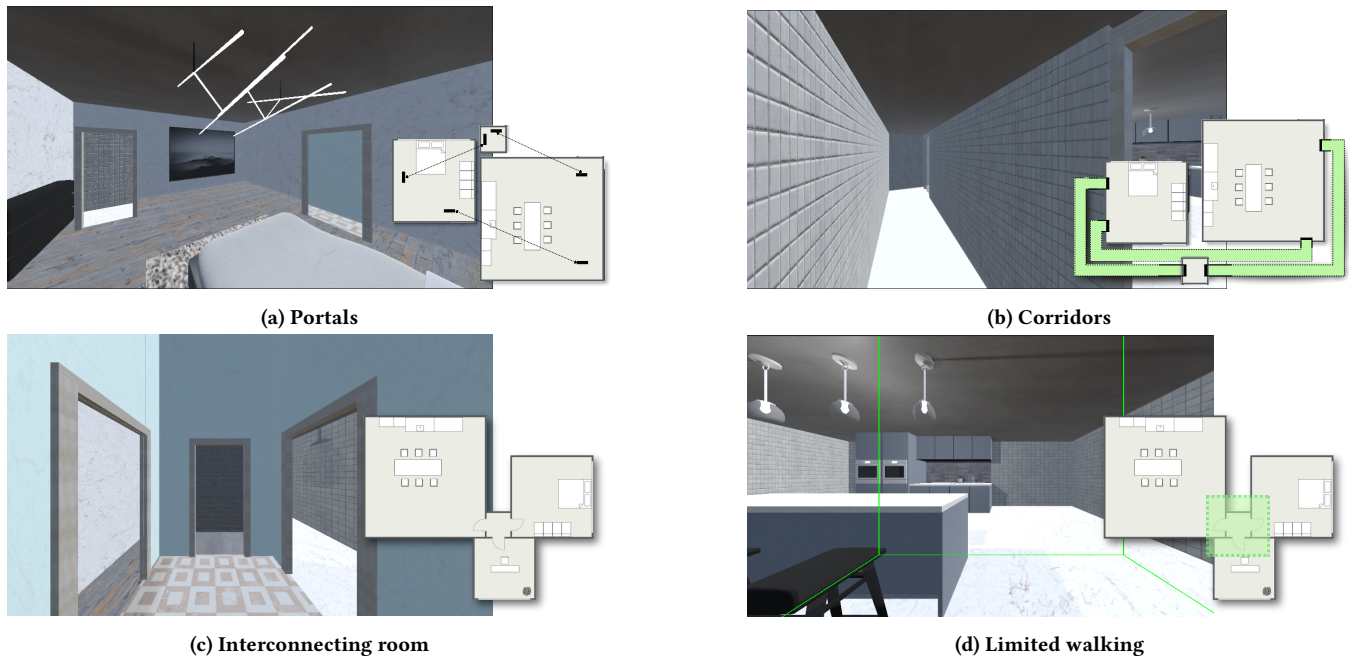


Figure 3: The four strategies used to connect the rooms in the virtual apartment.

Portals: Dynamic gateways that link separate and often incongruent spaces. These gateways facilitate instantaneous transitions between different settings, allowing for a seamless navigation. Portals can be visually conspicuous or subtly integrated, depending on the narrative context.

Corridors: Create the impression of greater distances within a confined physical environment, fostering a sense of exploration, as the corridor serves as both a tangible and psychological pathway to novel spaces. They facilitate smooth transitions between areas, subtly incorporating changes that cue users about the upcoming space. This approach offers an effective alternative to redirected walking techniques, which use rotation gains to avoid collisions with physical boundaries. Corridors naturally guide users back to the play area's center.

Interconnecting Room: A central hub that acts as a navigational anchor, simplifying spatial orientation in a complex network of rooms. Each room, connected back to the hub, allowing users to explore divergent spaces while maintaining a coherent spatial reference point. While it can alter the original layout and spatial perception, this approach is valuable in settings such as museums, where the exact layout is less crucial, and quick access to different rooms is beneficial.

Limited Walking: This approach does not adjust the user's stride and maintain the original layout, resulting in some rooms being too large for the limited physical space. When the user nears the play area limits, a safety grid appears. This method is particularly suitable for users sensitive to cybersickness, as it mimics real walking without stride alteration and is often the only feasible option for those who experience discomfort in modified virtual environments.

4 INTERVIEWS AND FEEDBACK

We conducted interviews with 10 experts across different fields of HCI to validate the efficacy and feasibility of our proposed solution and gather feedback for further enhancements. This process aimed to enrich our solution by considering diverse suggestions and viewpoints, striving to achieve a robust solution capable of accommodating an extensive range of scenarios and addressing diverse user restrictions.

The interviews were designed as semi-structured sessions to allow participants the freedom to direct the conversation according to their expertise and insights. However, to ensure a well-rounded discussion, specific topics were consistently addressed in each interview:

- **Application scenarios:** Participants were encouraged to share their perspectives on potential real-world applications where our proposed solution could be effectively implemented.
- **Usability challenges:** Discussions centered around identifying usability challenges or limitations faced by users, including the exploration of potential discomforts, restrictions in movement, and challenges in interaction.
- **Enhancing user experience:** Participants were prompted to provide feedback aimed at enhancing the overall user experience within constrained VR settings. This encompassed suggestions for improving navigation, mitigating motion sickness, and optimising user comfort.
- **Refinement suggestions:** Participants were encouraged to offer suggestions for refining the tool and overcome limitations identified during the discussions.

The interviews began with a short 3-minute pitch, briefing participants about the problem statement and our proposed solution. The intention was to avoid biasing the participants by delving into extensive details that might influence their initial perceptions. Following this, we collected initial feedback to gauge participants' immediate thoughts and observations. Subsequently, participants engaged in a hands-on experience with the developed system. They were presented with the apartment case study described in section 3.3. After the hands-on experience, participants were encouraged to share detailed feedback, insights, and observations based on their interaction with the system and the scenarios presented. This post-experience feedback aimed to capture their perceptions after engaging with the solution firsthand. Individual interviews, lasting approximately 30 to 45 minutes, were conducted in person to ensure that all participants had the opportunity to experience the VR demo firsthand.

4.1 Results

All the insights gathered during the interviews are summarised in Table 1. Participants have identified various application scenarios that showcase the versatility of our solution. These scenarios include virtual experiences such as tours, museums, and galleries, accessible just by walking through one's living room. For instance, users could virtually visit museums or explore potential homes,

offering enriching experiences beyond geographic limits. The system also holds promise in interactive and collaborative environments, ranging from enhancing e-commerce platforms with engaging shopping experiences to creating virtual spaces for remote collaboration. Moreover, this tool can also be applied in simulation and training for emergency scenarios. Schools and companies could use it for immersive emergency response training, while firefighters could train in random generated virtual scenarios for better preparedness.

These applications come with their own set of usability challenges. Spatial configuration and boundaries present significant issues, requiring the system to accommodate a variety of physical spaces, including those with irregular shapes, while also maintaining distance perception in virtual environments. The placement of portals must be strategic to prevent alignment with physical obstacles. User accessibility and comfort are crucial as well; the system must cater to individuals with limited mobility, address the issue of claustrophobia in confined spaces, and reduce simulator sickness. Additionally, environmental and external factors, such as changes in the play area or the presence of multiple users can lead to unexpected objects entering the play area or potential collisions in spaces shared by several users.

For suggestions and improvements, some suggested that portal design should be contextually appropriate, with aesthetics varying to suit different scenarios — for example, using traditional doors

Table 1: Overview of the feedback collected during the interviews with HCI researchers

Application Scenarios		
Virtual Experiences: - Virtual tours - Virtual museums and galleries - Exploration of environments - Real estate visualization	Interactive and Collaborative Environments: - E-commerce platforms - Collaborative spaces for remote teamwork - Gaming - Virtual Escape Rooms - Chat Rooms	Simulation and Training: - Construction testing (e.g. amusement parks) - Training for emergency scenarios
Usability Challenges		
Spatial Configuration and Boundaries: - Determine minimum viable physical area - Irregularly shaped physical spaces - Uneven terrain (e.g. sloping terrain) - Portal positioning - Abrupt transition between rooms - Fixed play area obstacles (e.g., a pillar) - Preserving distance perception	User Accessibility and Comfort: - Accessibility for people with limited mobility - Addressing claustrophobia in confined spaces - Addressing simulator sickness for sensitive users	Environmental and External Influences: - Play area changes during experiment - Multiple users in the same play area - Space limitations affecting activity types - Consistency in environmental sounds
Enhancing User Experience and Refinement Suggestions		
Portal Design and Functionality: - Context-orientated portal aesthetics - Rotating portals based on user orientation - Show both original doors and added portals	Navigation and Transition Enhancements: - Room transition with gradual stride adjustment - Corridors for smoother room transitions - Real-time virtual stride adjustments - Adjust space before or during the experience - Provide maps for user orientation	User Experience and Adjustments: - FOV adjustment based on walking speed - Alerts for changes in the physical playing area - Include demos for new users - Customizable soundscapes - Dynamic environmental lighting - Moving objects within VR (e.g., move a chair) - Algorithmic suggestions for input parameters - Space generation and adjustment in real time - Consideration of proxemic relationships in VR

in real estate applications and futuristic portals in gaming environments. The ability of portals to rotate and adapt to the user's orientation can significantly enhance navigation, especially in complex layouts such as large galleries. In terms of navigation and transitions within virtual spaces, implementing gradual stride adjustments and incorporating virtual corridors can help in environments with varying room sizes, such as in virtual real estate tours. Real-time adjustments, including door repositioning, provide flexible customization to correct or fine-tune details, ensuring the final virtual environment is as tailored as possible to the scenario and the end user. Modifying the field of view according to walking speed can help alleviate motion discomfort. Additionally, implementing alerts for changes in the physical environment is crucial for safety, especially in shared spaces. Finally, personalized soundscapes and dynamic ambient lighting aim to deepen immersion, reflect real-world conditions, and smooth the transition between the virtual and the real world.

4.2 Discussion

The interviews confirm the significant potential of our VR tool in enhancing user engagement across various domains, including virtual tours, e-commerce, training environments, museums, galleries, and remote collaborative work. Its unique feature of enabling users to simply walk within their small physical play area distinguishes it from traditional handheld controller-based systems, offering a more intuitive and immersive user experience. The presented use case is just one example illustrating how this tool can condense a virtual environment of approximately 100 m² into a walkable space within a 6.25 m² play area.

Our work aligns with the existing literature on the role of VR in transforming user experiences but takes a step further by concentrating on the adaptability of virtual environments and the natural navigation feature in small physical spaces. This focus is key for designing experiences tailored for home use, thereby reaching more users by demonstrating them that VR is becoming more accessible, enabling common people to enjoy immersive experiences anywhere without the need for a large room. In practical applications, our system can improve training methods in emergency services, offering realistic simulations. For the commercial sector, including retail and real estate, it introduces a new dimension in customer engagement, creating more accessible solutions for those who cannot reach physical stores and facilitating remote interactions while saving time for both clients and agents.

Feedback from researchers has been instrumental at this early stage of development in highlighting usability challenges and safety concerns, particularly in mapping complex environments and ensuring safety in shared or confined spaces. This feedback paves the way for further research and extensive user testing across diverse demographics.

This work demonstrates the potential of our VR tool in creating immersive and adaptable experiences, underscored by its natural navigation feature. Despite challenges, the system shows significant promise in enhancing various aspects of life, both daily and professional. This feedback from HCI researchers validates the value of our solution and sets the stage for the next phase of development,

focusing on user-centric improvements to create a more robust solution.

5 CONCLUSION

In conclusion, this study introduces a VR tool currently in development that enables walking navigation in limited physical spaces, allowing adjustments to virtual environments to suit specific contexts and end users. Employing non-Euclidean geometry and procedural generation concepts, our approach enable walking within confined spaces, distinguishes it from traditional VR systems reliant on handheld controllers. This tool opens up possibilities for more natural navigation in virtual tours, e-commerce, training simulations, and remote collaborative work. We presented a use case involving virtual apartment tours to gather feedback from HCI researchers, confirming the tool's great potential to overcome space constraints and enhance user engagement in various domains. While challenges in usability, safety, and environment mapping persist, the positive reception and constructive criticism obtained in the interviews highlight the significant potential of this tool in both everyday and professional contexts, justifying further research.

6 FUTURE WORK

The future development of our VR solution will concentrate on key areas, initially prioritizing the enhancement of the system's mapping capabilities for diverse physical environments. Additionally, we plan to implement real-time environmental alerts to address user safety concerns, preventing potential hazards such as collisions, particularly for users with limited mobility or susceptibility to motion sickness. Prioritizing extensive user testing across diverse demographics is imperative to better understand and cater to various user needs, enhancing the inclusivity and accessibility of our VR solution. Furthermore, we aim to study VR environments for specific applications beyond the real estate use case, such as virtual museums and galleries, to augment the solution's relevance and effectiveness in different fields. Our short-term focus also involves exploring longitudinal studies to evaluate the long-term effects of VR on users' well-being and its effectiveness in various settings. These efforts are geared toward advancing the technical and user-experience aspects of our VR tool, with the ultimate goal of solidifying its position as a versatile, user-centric solution in virtual experiences.

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